NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

# Course Name: ETHICAL HACKING

## Assignment- Week 1

### TYPE OF QUESTION: MCQ/MSQ/SA

**Number of questions**: 10 **Total mark: 10 x 1 = 10**

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**QUESTION 1:**

Which of the following statements best describes steganography?

a. Encrypting information using public and private keys.
b. Embedding information within another cover medium.
c. Tunnelling data using HTTPS and TLS.
d. Compressing data to reduce its transmission size.
e. None of these.

**Correct Answer: b**

**Detailed Solution:** Steganography is the practice of hiding the existence of a message by embedding it within another medium such as an image, audio, video, or executable file. It is distinct from encryption (which makes data illegible but doesn't hide its existence), does not eliminate the need for secrecy in all cases, and is unrelated to data compression.
Thus, the correct option is (b).

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**QUESTION 2:**

Which of the following are valid examples of steganography?

a. Hiding text data within an image file.
b. Hiding secret data within an audio clip.
c. Encrypting a file so only the recipient can decrypt it.
d. Hiding data within an executable file.
e. Digitally signing a document to prove authorship.
f. None of these.

**Correct Answer: a, b, d**

**Detailed Solution:** Steganography involves hiding data within cover media like images, audio, or executables. Encrypting a file (option c) is encryption, not steganography, and digitally signing a document (option e) relates to authentication/non-repudiation, not information hiding. Thus, the correct options are (a), (b), and (d).

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**QUESTION 3:**

Which of the following is/are examples of behavioural biometrics?

a. Fingerprint
b. Signature dynamics
c. Gait
d. Iris pattern
e. Keystroke dynamics
f. Retina pattern
g. Facial structure

**Correct Answer: b, c, e**

**Detailed Solution:** Behavioural biometrics relate to patterns in human behaviour, such as signature dynamics, gait, and keystroke dynamics. Fingerprint, iris pattern, retina pattern, and facial bone structure are all physiological (physical body-based) biometrics, not behavioural. Thus, the correct options are (b), (c), and (e).

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**QUESTION 4:**

Which biometric modality is generally considered to offer very high uniqueness and reliability for identification?

a. Voice recognition
b. Facial recognition
c. Iris scan
d. Signature dynamics
e. Gait analysis

**Correct Answer: c**

**Detailed Solution:** Iris recognition offers very high uniqueness and reliability because the iris pattern is highly distinctive for everyone. Thus, iris scanning is a highly reliable biometric modality for identification.
Thus, the correct option is (c)

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**QUESTION 5:**

What is a Denial-of-Service (DoS) attack?

a. An attack that deletes or modifies files stored on a system.
b. An attack that prevents legitimate users from accessing a service.
c. An attack that specifically targets and destroys stored password databases.
d. None of these.

**Correct Answer: b**

**Detailed Solution:** A Denial-of-Service attack aims to make a service or system inaccessible to its legitimate users, typically by overwhelming its resources, rather than directly deleting files or targeting password databases.

Thus, the correct option is (b).

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**QUESTION 6:**

How does a Smurf DoS attack typically work?

a. It sends oversized IP packets to crash the target host.
b. It sends forged ICMP Echo requests to a network's broadcast address with the victim's spoofed source IP, causing many hosts to flood the victim with replies.
c. It exploits buffer overflow vulnerability in web server software.
d. It uses SQL injections to compromise a backend database.
e. None of these.

**Correct Answer: b**

**Detailed Solution:** In a Smurf attack, the attacker sends ICMP Echo requests to a network's broadcast address with the source IP spoofed to the victim's address. All hosts on that network then send Echo replies to the victim, overwhelming it with traffic. Oversized packets describe a Ping of Death attack, not a Smurf attack.

Thus, the correct option is (b).

**QUESTION 7:**

Which of the following attacks exploits the TCP three-way handshake by leaving connections in a half-open state?

a. Ping of Death attack
b. SYN flooding attack
c. Smurf attack
d. DNS spoofing attack

**Correct Answer: b**

**Detailed Solution:** SYN flooding exploits the TCP handshake by sending numerous SYN requests without completing the handshake (never sending the final ACK), leaving many half-open connections that consume server resources and prevent legitimate connections.

Thus, the correct option is (b).

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**QUESTION 8:**

What does PGP (Pretty Good Privacy) primarily provide for secure email communication?

a. Authentication of the sender
b. Confidentiality of the message content
c. Guaranteed message availability
d. Protection against email spam filtering

**Correct Answer: a, b**

**Detailed Solution:** PGP provides authentication and confidentiality using a combination of hashing and encryption (both symmetric and asymmetric). It does not guarantee message availability, guaranteed delivery, or spam protection.

Thus, the correct options are (a) and (b).

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**QUESTION 9:**

What is the primary purpose of the Domain Name System (DNS)?

a. To provide end-to-end encryption for email communication.
b. To translate human-readable domain names into IP addresses.
c. To compress and fragment large network messages.
d. To hide secret data inside cover media files.
e. None of these.

**Correct Answer: b**

**Detailed Solution:** DNS's core function is to resolve human-readable domain names (such as www.example.com) into their corresponding numeric IP addresses, allowing users to access resources without memorizing IP addresses.

Thus, the correct option is (b).

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**QUESTION 10:**

Consider a grayscale image of size 1500 x 1200, where each pixel is stored using 8 bits. Using LSB steganography (1 bit hidden per pixel), how many bytes of information can be hidden in this image?

**Correct Answer: 225000**

**Detailed Solution:** Total pixels = 1500 x 1200 = 1,800,000. Since each pixel allows hiding 1-bit, total hidden bits = 1,800,000 bits. Converting to bytes: 1,800,000 / 8 = 225,000 bytes.
Thus, the correct answer is 225000.

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